

7	(a) (i)	2 protons and 2 neutrons	B1	[1]
	(ii)	e.g. positively charged $2e$ mass $4u$ constant energy absorbed by thin paper <i>or</i> few cm of air ($3\text{ cm} \rightarrow 8\text{ cm}$) <i>(not low penetration)</i> <u>highly</u> ionizing deflected in electric/magnetic fields <i>(One mark for each property, max 2)</i>	B2	[2]
	(b)	<u>mass-energy</u> is conserved difference in mass 'changed' into a form of energy energy in the form of kinetic energy of the products / γ -radiation photons / e.m. radiation	B1 B1 B1	[3]